

BMW Group Standard

Released

GS 90018-2
2022-04

Replacement for
GS 90018-2:2018-10

Requalification of product and process at suppliers

Process, checklist

Edited by

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The user of this document is under obligation to verify its current validity.

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Foreword

The German version is binding.

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This Group Standard has been coordinated with the responsible departments of the BMW Group. The names of the persons creating, releasing and checking this Group Standard are filed and can be retrieved in the master data of the TEREK system.

Amendments

The following amendments have been made to GS 90018-2:2018-10:

- section 3 and section 6 changed;
- Table 4 updated: "Evaluation" of rating "red";
- section 7 changed:
 - added positions: 6.13, 7.19, 7.20;
 - following positions revised: 1.1, 1.7, 2.2, 2.4, 2.6, 5.3, 5.7, 5.9, 5.10, 6.8, 7.13, 7.14;
 - column „Relevant“ removed from check list;
- normative references updated;
- standard editorially revised.

Previous editions

2006-02, 2007-10, 2010-09, 2013-10, 2017-01, 2017-08, 2018-09, 2018-10

1 Scope and purpose

This Group Standard defines the process and the contents of requalifications. General specifications are defined in GS 90018-1.

It applies to all supplied parts of the BMW Group as of start of production.

2 Normative references

This standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

DIN 55350-11	Concepts for quality management; Part 11: Supplement to DIN EN ISO 9000:2005
GS 90000-1	Technical regulations; Terms, structure
GS 90017	Continuous assurance of product reliability in the series phase; Process description
GS 90018-1	Requalification of product and process at suppliers; General specifications
GS 90036	Identification of parts; Rules for definition and implementation of traceability of parts
GS 91007-1	Measurement plan process for trim and final parts; Fundamentals, process description
GS 91007-2	Measurement plan process for trim and final parts; Integrated intersecting plane model (ISL)
GS 91007-3	Measurement plan process for trim and final parts; Format and contents of measurement plan
GS 91007-4	Measurement plan process for trim and final parts; Measurement report
GS 91008	Funktionsmaßkonzept (FMK); Process, documentation, terms
GS 95009	Electrostatics; Protective measures against electrostatic discharge (ESD)
GS 95015	Components in motor vehicles; Requirements placed on the supplier's defect analysis and elimination for zero kilometer issues
GS 95024-1	Electrical and electronic components in motor vehicles, general requirements
GS 95024-2	Electrical and electronic components in motor vehicles, electrical requirements and tests in 12-V onboard electrical systems
GS 95024-3-1	Electrical and electronic components in motor vehicles, environmental requirements and tests
GS 95024-3-3	Electrical and electronic components in motor vehicles, run-in of components
GS 98000	Statistical capability studies
IATF 16949:2016	Quality management system requirements for automotive production and relevant service parts organisations
QV 11111	Appropriate technical cleanliness of components and fluids in BMW power train; Inspection and strategies; for design, manufacturing in series production and rework, replacement and spares in service
VDA Volume 19 Part 2	Quality management in the automotive industry; Technical cleanliness in assembly; Environment, logistics, personnel and assembly equipment

3 Abbreviations, terms and definitions

3.1 Abbreviations

CAQ	Computer Aided Quality
CCC	China compulsory certificate
COP	Conformity of production
BTRV	Component tracking
ESD	Electrostatic discharge
FIFO	First in - first out
FMEA	Failure method evaluation analysis
FMK	Functional dimension concept
GIS	Global Information Services
GWK	Warranty and goodwill
INMETRO	National Institute of Metrology Standardization and Industrial Quality
KFG	No fault found
LDB	Supplier database
LPKM	Supplier-performance and competence management
OEE	Overall Equipment Effectiveness
OFK	Upper management executive
PLP	Control plan
PQM	Product Quality Management
QM	Quality management
QMT	Quality management parts
SPC	Statistic process control
TecSa	Technical cleanliness
TPb	Technical product specification
ZUV	Reliability

3.2 Terms

8D report	Definition according to GS 95015
Special Characteristics	Product characteristics or manufacturing process characteristics having an effect on safety or compliance with official regulations, fit, function, performance or the further processing of the product.
Measurement plan	Additionally to the release model the measurement plan serves for the coordination and description for customer-relevant requirements (internal/external) with regard to the geometric quality of the component (e.g. GS 91007-1 to GS 91007-4).
Production control plan	Definition according to IATF 16949:2016
Test plan	Definition in accordance with DIN 55350-11
Technical product description	Definition in accordance with GS 90000-1
Reliability indicator	Definition in accordance with GS 90017
Reliability characteristic	Quality characteristic that characterizes the response of the unit during or after specified periods of time at predetermined application conditions (following DIN 55350-11).
Technical cleanliness	Definition in accordance with QV 11111

3.3 Definitions

There are no definitions in this document.

4 Designation

There is no designation in this document.

5 Process

5.1 Phases of requalification inspection

A requalification inspection consists of

- a preparation phase (see section 5.2),
- an execution phase (see sections 5.3 and 5.4) and
- a post-processing phase (see section 5.5).

5.2 Preparation

As part of the preparation phase, the process owner (see GS 90018-1) shall compile all available information for the scope to be requalified, based on the questions defined in Table 1.

Based on the present and other information, the questions relevant for the requalification shall be determined by the process owner. These questions shall be marked in the checklist (see section 7).

Table 1: Questions concerning preparation

Reference to checklist	Question / content
1.1 / 4.2 / 6.5	Check release and drawing status as well as measurement plan.
3.6 / 3.7	Has a reliability assurance for the component been agreed upon in the series in accordance with GS 90017?
4.1	Requalification planning: Are current plans for the requalification of product and process in accordance with GS 90018-1 and GS 90018-2 available?

Table 2: Recommendation for preparation

Format	Recommendation of documents to be prepared
Printout	Drawing Requirements specification Quality agreement Production control plan Production process flow chart/-layout Measurement plan
IT	Relevant BMW Group Standards Status initial sample inspection Parts history Process FMEA Measurement results Working instructions CAQ-system Quality objective and quality reporting Audit results incl. measures Product requalification planning Process requalification planning Complaint processing customer Complaint processing sub-supplier Supplier management system Process parameters System maintenance and Service Gauge capability documentation Inspection and test equipment management Qualification and planning status

5.3 Execution

The general specifications concerning the execution of requalification are described in GS 90018-1. A guide for the scheduling is illustrated in Table 3.

Table 3: Execution

Duration	Activity
approx. 0,5 days	Preparation
0,7 days	Discussion of the topics and viewing of the documents
1 day	Process examination: – Product requirements – Production scheduling – Measurement and validation – Requalification and auditing – Objectives and control loops – Sub-supplier management – Shop floor – Maintenance – Test equipment
0,3 days	Summary, evaluation and final discussion

5.4 Assessment of results

As part of the execution phase, the answers to the questions defined in the checklist shall be compiled and evaluated in accordance with the traffic light principle (see Table 4). The assessment shall be summarized for the individual question groups and entered in the evaluation cover sheet (see section 6).

The overall assessment is derived from the individual assessments of the question groups and shall also be recorded in the evaluation cover sheet.

The individual results shall be merged such that the poorest result of a question group is decisive for the overall assessment of this question group. For example, if a question group is assessed 5 times with "yellow" and only once with "red", the overall assessment for this group will be "red".

The assessment of the individual question groups shall be merged in the same way in order to provide the overall requalification result.

Table 4: Assessment of results

	Assessment	Possible call for action
red	– Critical parts supply – Unacceptable overall situation – Safety and legally relevant defects – Essential requirements are not fulfilled – Serious delays compared to target schedule can be observed – Even with special measures, schedule cannot be met – Considerable exceedance of the limit values and/or target values – Field complaints can be expected – Complaints from the BMW manufacturing plants can be expected – Characteristics relevant for the function are not validated/safeguarded or only validated/safeguarded inadequately – Quality problems cannot be solved immediately – Risks from deficiencies cannot be evaluated sufficiently – Plan of measures by supplier for defect elimination is not available and/or not effective	– Initiate immediate change of the situation – Take measures to avoid faults/defects in series production – Immediate incorporation / information of the affected BMW departments – Define special measures – Make a review appointment – Fundamental decision required in the project (e.g. postponement of schedule) – Exclude safety risks – Carry out escalation (see Figure 2) – Repeat requalification
yellow	– Supply capability risk discernible – Does not correspond to the BMW requirement to the full extent – Planning deviations (not within schedule / cost plan) – Requirements not completely fulfilled – Risk existing and described – Little exceedance of limit and/or target values	– Measures are required – Strive for improvements – Gradual modifications are sufficient (no immediate measures are required) – Initiate improvement of processes – Define effective remedial measures – Schedule resubmission in committees – Show consequences
green	– Parts supply is ensured – Corresponds to the BMW objective to the full extent – Complies with all requirements – No essential potential for improvement – None and/or only few deviations from the plan – Quality losses are not expected	– No measures required – No additional measures required that trigger expenditure – Further improvements, the earliest, at the successor model

5.5 Revision/Monitoring

For each question of the check list evaluated with "yellow" or "red", as illustrated in section 7, the systematic causes shall be determined , measures agreed upon, process owners assigned and schedules defined.

The measures plan (see section 8) shall be shown to the costumer within 14 calendar days after the last requalification day.

The measures shall be implemented within a period of 3 months after the requalification and they shall be confirmed as effective within one additional month.

Proof of effectiveness may be substantiated during a review meeting. In the case of a red overall result the requalification shall be repeated. The BMW Group may also appoint a review meeting due to other reasons (Figure 1).

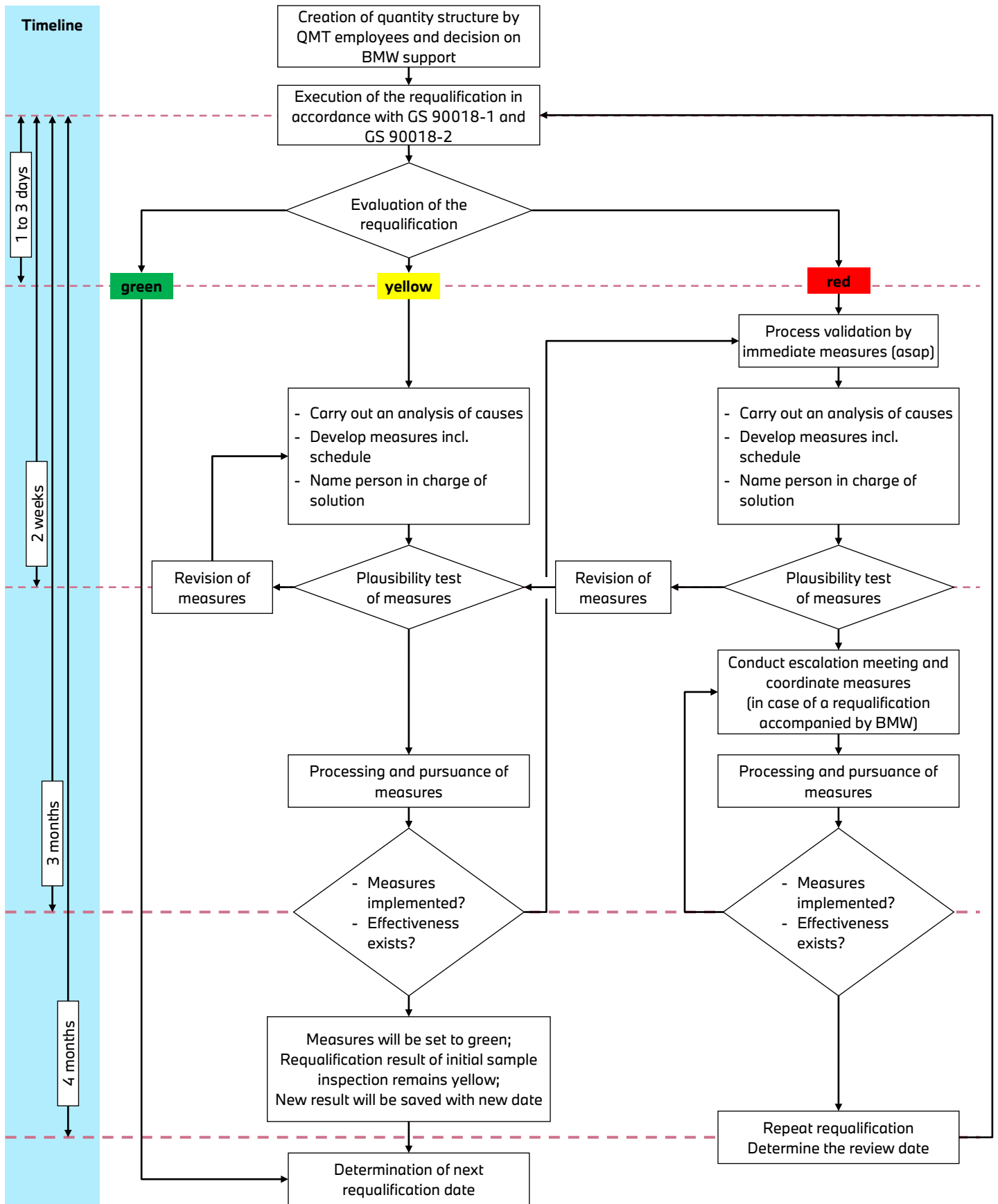


Figure 1: Procedure plan revision/monitoring

In case BMW accompanies the requalification at the supplier, Figure 2 illustrates the procedure of a possibly necessary escalation in the event of a red and/or yellow overall evaluation. The respective QMT employee of the

BMW Group is responsible for the conduction of the escalation.

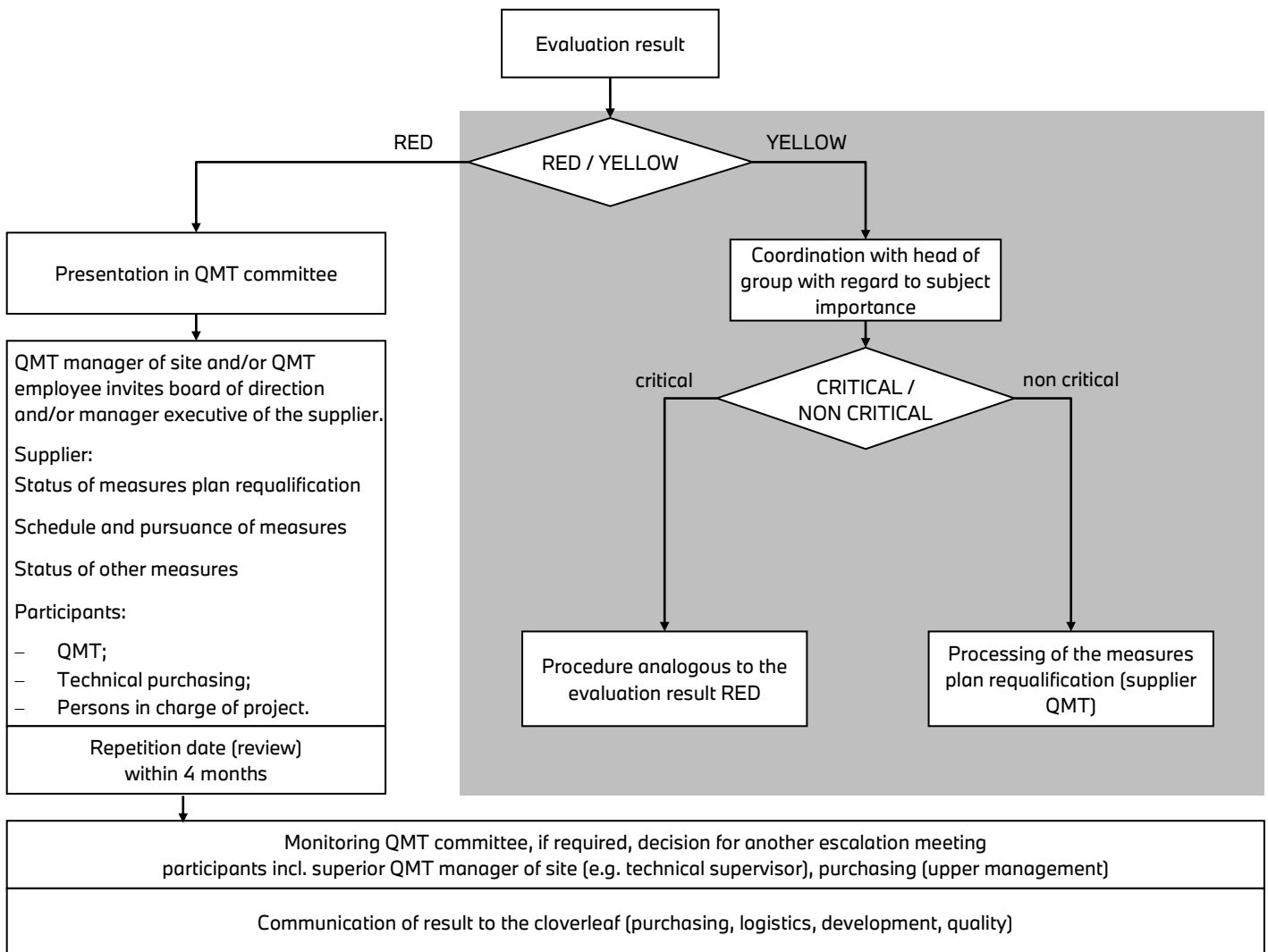


Figure 2: Procedure of escalation

6 Evaluation cover sheet



ROLLS-ROYCE
MOTOR CARS LTD

Evaluation cover sheet	
Requalification product and process in accordance with GS 90018-2	Date:

Project:	Supplier:
Description:	Production interval:
Part number:	No. of items:

Participant BMW QMT: _____ Logistics: _____ Purchasing: _____ Development: _____ Testing: _____ Other: _____	Participant supplier: _____ _____ _____ _____ _____
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Pos.	Checkpoints	Evaluation			Note
		RED	YELLOW	GREEN	
1	Product requirements	☐	☐	☐	
2	Production scheduling	☐	☐	☐	
3	Measurement and validation	☐	☐	☐	
4	Requalification and auditing	☐	☐	☐	
5	Objectives and control loops	☐	☐	☐	
6	Sub-supplier management	☐	☐	☐	
7	Shop floor	☐	☐	☐	
8	Maintenance	☐	☐	☐	
9	Test equipment	☐	☐	☐	
Evaluation sum					Out of 84 questions marked as "not relevant".
					Out of 84 questions still not evaluated.

Total evaluation	☐	☐	☐	Failed
				Conditionally passed
				Passed

Comment	
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Signatures
The traffic light evaluation is performed in accordance with GS 90018-2, section 5.4

7 Checklist

No. P T	Question	Explanations	Comment	Evaluation			Not relevant
				RED	YELLOW	GREEN	
1	Product requirements						
1.1 T	Requirements on product and process: Are all the required specifications of the customer completely available and are they current?	Requirements Specifications, drawings, BMW standards, Group Standards, TPbs, QMT requirements, scopes that require a product certificate (e.g. CCC, INMETRO), component tracking (BTRV)		☐	☐	☐	☒
1.2 T	Special characteristics: Are the special characteristics for the final products and components specified and updated?	Requirements Specifications, TPbs, inspection plan at the supplier (also reliability characteristics and technical cleanliness)		☐	☐	☐	☒
1.3 T	Initial sample inspection: Has an initial sampling inspection been performed for the current status of all supply scopes?			☐	☐	☐	☒
1.4 T	Part life histories: Are the part life histories for all delivery scopes and components current and complete?			☐	☐	☐	☒
1.5 T	Change management: Are the consequences of all changes (product and process) be evaluated (verified and validated) and are they coordinated with the costumer?	See IATF 16949:2016 section 8.5.6.1 "Monitoring of changes"		☐	☐	☐	☒
1.6 T	Product requalification: Have requalification tests been agreed with BMW and carried out?	See GS 95024-1, GS 95024-2, GS 95024-3-1, GS 95024-3-3 and GS 90018		☐	☐	☐	☒
1.7 P T	Marking of changes: Is the process for marking of changes known and how is the implementation ensured?	Consider AI based on data sheet and hardware		☐	☐	☐	☒
1.8 T	E/E component change during series production phase: Have qualification tests / inspections been performed for these components and what were the results?			☐	☐	☐	☒

No.	Question	Explanations	Comment	Evaluation			Not relevant
				RED	YELLOW	GREEN	
2	Production scheduling						
2.1 T	Control plan: Are intervals specified for regular checking of the characteristics of the components / final products / tools?	In production control plan in accordance with IATF 16949:2016		☐	☐	☐	☒
2.2 T	Control plan: Is the creation and updating of control plans supported by an effective IT solution?	Are fully updated control plans (e.g. response plans, emergency plans) available?		☐	☐	☐	☒
2.3 T	Control plan: Have the changes been documented and released by BMW since the process series?	It shall also be taken into account that the temporary changes (stand-by), the requirements for backup or alternative methods for process control are defined and anchored in the PLP.		☐	☐	☐	☒
2.4 T	Control plan: Does the production process plan / layout reflect the latest interrelation of product, process and tool supply in terms of time and space?			☐	☐	☐	☒
2.5 T	E/E further development: Is the 10-day flash concept implemented?			☐	☐	☐	☒
2.6 T	E/E component change during series production phase: Is it ensured that a process description is available for the use of new or announced components?			☐	☐	☐	☒
2.7 T	E/E further development: Is further development taking place and is the current I-stage plan available?			☐	☐	☐	☒
2.8 P	E/E further development: Is the production ESD-compliant in accordance with GS 95009?			☐	☐	☐	☒
2.9 P	E/E further development: Are the tests conducted in the overall system corresponding to the current planning status?			☐	☐	☐	☒

No.	Question	Explanations	Comment	Evaluation			Not relevant
				RED	YELLOW	GREEN	
3	Measurement and validation						
3.1 T	Functional dimension concept: Does a functional dimension concept (FMK) exist for the assembly (ZB) and for all relevant parts and is it integrated in the measurement plan?	Application of GS 91008		☐	☐	☐	☒
3.2 P T	Measurement planning: Does a measurement plan, agreed with BMW, exist and are the measurement reports compiled corresponding to the current measurement plan?	Drawing characteristics (functional dimensions, alignment systems), application of GS 91007-1, GS 91007-2, GS 91007-3 and GS 91007-4		☐	☐	☐	☒
3.3 T	Process capability: Is the long-term process capability proven for the special characteristics?	$C_{pk} \geq 1,33$ (also reliability characteristics)		☐	☐	☐	☒
3.4 P T	Special characteristics: Are the special characteristics for the final products and components checked and evaluated and are appropriate corrective measures taken in case of deviations?	Documentation of actual values (also reliability characteristics and technical cleanliness)		☐	☐	☐	☒
3.5 P T	Process assurance: Is safeguarding achieved in case of incapable processes?	e.g. 100 %-test, type, scope, documentation		☐	☐	☐	☒
3.6 P T	Reliability assurance: Are the reliability indicators / reliability tests determined, installed and are they subject to a monitoring?	Only for components for which a reliability validation is agreed in the series production in accordance with GS 90017, reliability indicators in accordance with GS 90017, inspection plan at the supplier with testing frequency and action limits		☐	☐	☐	☒
3.7 P	Reliability assurance: Are the reliability tests conducted and monitored at the end product and will deviations lead to an immediate process correction?	Only for components for which a reliability validation is agreed in the series production in accordance with GS 90017, endurance test (stress test), quick corrective loop according to GS 90017 . Is the process properly documented?		☐	☐	☐	☒

No.	Question	Explanations	Comment	Evaluation			Not relevant
				RED	YELLOW	GREEN	
3.8 P	Measurement reports: Is the informative value of the measurement reports sufficient for the final products and individual components?	Layout of the measuring reports, scope and frequency of the measurements (incl. special characteristics). Unambiguity with regard to e.g. tolerances, evaluation, key figures for process capability, change index assignment, production batch		☐	☐	☐	☒
3.9 T	Measurement reports: Are the test results recorded mandatory and systematically evaluated in a functioning IT solution?			☐	☐	☐	☒
3.10 T	Fail-safe: Is a determined process for "application of methods for fail-safe" documented?	"Fail-safe" IATF 16949:2016 section 10.2.4		☐	☐	☐	☒
3.11 P	E/E further development: Is the testing concept consistent and implemented in full?			☐	☐	☐	☒

No.	Question	Explanations	Comment	Evaluation			Not relevant
				RED	YELLOW	GREEN	
4	Requalification and auditing						
4.1 P T	Requalification planning: Are current plans for the requalification of product and process according to GS 90018 available?	Planning for requalification is compiled in PLP		☐	☐	☐	☒
4.2 T	Requirements on product and process: Was the requalification process carried out according to the plan and were the specified criteria met?	e.g. Maintenance plans, production control plan, special characteristics, emergency planning (IATF 16949:2016 section 6.1.2.3)		☐	☐	☐	☒
4.3 P T	Requirements on product and process: Do the parts comply with the requirements of the specified requalification tests of the product?	Complete dimensional and functional test taking into account the BMW specifications that shall be applied (e.g. drawing, TPb)		☐	☐	☐	☒
4.4 T	Audit planning: Are the audits conducted according to the audit plan?	Provide results from system audit, process audit and product audit; technical cleanliness audit according to VDA Volume 19 Part 2, QV 11111		☐	☐	☐	☒
4.5 T	Control plan: Are the special characteristics (incl. COP requirements / characteristics) taken into account in the requalification tests of the product?	Special characteristics (also reliability characteristics and technical cleanliness) in the production control plan in accordance with IATF 16949:2016		☐	☐	☐	☒
4.6 T	Personnel: Are there enough qualified employees available in the quality-relevant positions (quality-management, quality-planning, quality-validation, laboratory)?			☐	☐	☐	☒

No.	Question	Explanations	Comment	Evaluation			Not relevant
				RED	YELLOW	GREEN	
5	Objectives and control loops						
5.1 P T	Quality targets: Are the agreed quality-targets monitored?	What system is used to perform regular recording and monitoring? Which systems are used beyond the Partner Portal of the BMW Group?		☐	☐	☐	☐
5.2 P T	Quality targets: Are processes established to define and sustain measures in case of missed targets?			☐	☐	☐	☒
5.3 P T	Sustainability: Are the measures, deducted from the audits, implemented consistently and effective?	Analysis of causes, definition of measures and scheduling of measures, determination of responsibility, processing of measures and test of effectiveness		☐	☐	☐	☐
5.4 P	Sustainability: If measures were agreed upon from the process series / last requalification, are they implemented and effective?	e.g. Measures sheet - Process Series and Requalification		☐	☐	☐	☒
5.5 T	Internal rework / scrap: Are measures defined to meet increasing internal rework / scrap rates?			☐	☐	☐	☒
5.6 T	Process FMEA: Does a process FMEA exist and is it updated, if required?	Special product and process characteristics (also reliability characteristics) have to be marked in the FMEA		☐	☐	☐	☒
5.7 P	Treatment of faulty products: Is a procedure defined for handling faulty units the customer has complained about?	Corrective measure and proof of effectiveness, implementation also at similar processes and products (e.g. 8D-report, FMEA), field failures, 0 km-failures		☐	☐	☐	☒
5.8 T	Treatment of faulty products: Is the complaint process (internal) supported by a functioning IT solution?	IT solution for internal control and integration of supplied parts		☐	☐	☐	☒

No. P T	Question	Explanations	Comment	Evaluation			Not relevant
				RED	YELLOW	GREEN	
5.9 P	Fault correction: Are processes established to reduce problems occurring in the process chain?			☐	☐	☐	☒
5.10 P T	Traceability: Are agreements with regard to traceability of products complied with in accordance with GS 90036?	e.g. Individual plants / installations, employees, production data and times, shifts, batches (WBS/ASN) of supply parts; part identification (component tracking);		☐	☐	☐	☒
5.11 P T	Process disturbances: Are all process disturbances reported by BMW known and is their sustainable processing guaranteed?	Product audit BMW, quality at delivery, production process BMW, driving dynamics test, etc., assurance via 8D-report		☐	☐	☐	☒

No.	Question	Explanations	Comment	Evaluation			Not relevant
				RED	YELLOW	GREEN	
6	Sub-supplier management						
6.1 P T	QM system: Is it ensured that all sub-suppliers work in accordance with an established QM system?	Following IATF 16949:2016 section 8.4.2.3 "Supplier quality management system development" (e.g. distance, language barriers)		☐	☐	☐	☒
6.2 T	Supplier evaluation: Are the sub-suppliers evaluated with regard to quality of the delivered products, interferences at the costumer and delivery reliability?	See IATF 16949:2016 section 8.4.2.4 "Supplier monitoring", among others agreement of quality-objectives		☐	☐	☐	☒
6.3 T	Treatment of faulty products: Is a procedure in place for processing complaints regarding a sub-supplier?	e.g. 8D-report, fault elimination process agreed incl. processing times		☐	☐	☐	☒
6.4 T	Quality at delivery: Is the quality at delivery assured for supplied parts and materials?	e.g. Preventive quality-work, test of incoming goods, process audits at the sub-supplier Laboratory technique: Does the sub supplier possess the necessary means of assessment and analysis, such as a laboratory?		☐	☐	☐	☒
6.5 T	Requirements on product and process: Do the specifications of the costumer and the agreements with the sub-suppliers correspond to the current status?	e.g. Requirements specification, drawings, TPb, initial sample		☐	☐	☐	☒
6.6 T	Change management: Will the consequences of all changes (product and process) caused by the sub-supplier be evaluated (verified and validated) and will they be coordinated with the costumer?	See IATF 16949:2016 section 7.5.3 "Change control" part life history for the component, raw material, initial sample inspection, release process		☐	☐	☐	☒
6.7 T	Change management: Do all the parts to be processed (in-house and purchased parts) comply with the current change status?	e.g. State of drawings at BMW / supplier (change index), TPbs, deviations in production		☐	☐	☐	☒

No.	Question	Explanations	Comment	Evaluation			Not relevant
				RED	YELLOW	GREEN	
6.8 P T	Requalifications: Are regular requalifications performed for the products and processes of the sub-suppliers?	Is a requalification affecting sub-suppliers performed by the 1st tier supplier on a yearly basis?		☐	☐	☐	☒
6.9 T	Initial sample inspection: Has a procedure been defined regarding exceptions in case of missing/negative initial sampling inspections (ISIRs)?	e.g. Launch suitability assessment, special releases, 100 %-tests		☐	☐	☐	☒
6.10 T	Initial sample inspection: Is the current status of all scopes of supply by the sub-supplier to the supplier sampled initially?	Explanation of systematic recording of initial sampling (e.g. transparency, history), compliance with official regulations for procured products		☐	☐	☐	☒
6.11 T	Special characteristics: Are the special characteristics with regard to the product and the production process specified and agreed upon with the sub-supplier and are these subject to monitoring as required?	Documentation in Q-agreements (also reliability characteristics and technical cleanliness)		☐	☐	☐	☒
6.12 P T	FIFO: Is the adherence to FIFO assured in the supply stores?	Are obsolete products controlled in the same way as defective products (e.g. paints, adhesives)?		☐	☐	☐	☒
6.13 T	E/E Requirements on product and process: Is the genuineness of electrical/electronic components ensured when using brokerware?	e.g. Emergency plan (incl. obligation to notify BMW; can the genuineness of the products be confirmed?)		☐	☐	☐	☒

No.	Question	Explanations	Comment	Evaluation			Not relevant
				RED	YELLOW	GREEN	
7	Shop floor						
7.1 P	Control plan: Is series monitoring performed according to the specifications from the PLP?	e.g. Type, scope, frequency (see control plan)		☐	☐	☐	☒
7.2 P	Process parameters: Are all process parameters intended for monitoring actually monitored, recorded and documented in the required scope <u>at the site</u> as planned?	e.g. In overview tables for the possible parameters: pressure, temperature, velocity, time, position, intervention limits		☐	☐	☐	☒
7.3 P	Verification of set-up processes and verification following production stop: Is the release of the first unit / release after interruption / release of last unit carried out and documented?	The release of a production system before the begin of batch production and release after an interruption (e.g. breakage of tool) shall be stipulated in a binding manner; process plan		☐	☐	☐	☒
7.4 P	Cycle time: Is the agreed cycle time maintained?	See results of process series / launch curve / requalification, shift model		☐	☐	☐	☒
7.5 P	Working instructions: Are all required working instructions available at the workstation and can instructing be proven (complete, signed, with correct version)?	See working instruction and picturized specifications IATF 16949:2016 section 8.5.1.2 "Marking and Traceability"		☐	☐	☐	☒
7.6 P	Personnel: Is trained permanent staff installed in sufficient capacity and is this documented?	e.g. Personnel for manufacturing, tests capacity, training matrix, regulation for substitutes, job training plan, subcontracted worker quota		☐	☐	☐	☒
7.7 P	Identification of BMW property: Are tools / test devices belonging to BMW permanently identified?	Clear identification of property relation e.g. "Property of BMW"		☐	☐	☐	☒
7.8 P	Other operating resources and facilities: Are currently other facilities or operating resources used in the production process deviating to the released series production facilities (incl. operating supplies)?	Obligation to report		☐	☐	☐	☒

No.	Question	Explanations	Comment	Evaluation			Not relevant
				RED	YELLOW	GREEN	
7.9 P	Work environment: Does the work environment comply with the product and process requirements?	e.g. order, cleanliness, specific requirements to technical cleanliness, acoustics, disturbing factors, ergonomics, accessibility, lighting conditions		☐	☐	☐	☒
7.10 P	Provision of parts: Is the provision of parts in the production area implemented for the series process?	e.g. Racks, parts containers, tables, stores accessibility, available space with regard to maximum capacity, protection from contamination and environmental effects		☐	☐	☐	☒
7.11 P	Assurance against improper handling: Is improper handling prevented (e.g. Poka-Yoke)?	e.g. Tools, material		☐	☐	☐	☒
7.12 P	Assurance against improper handling: Does the provision of parts include sufficient precautions against improper handling?	Mixing up of parts impossible?		☐	☐	☐	☒
7.13 P	Product handling: Is the respective identification, handling, package, storage and corresponding protection of the products ensured throughout the complete process chain?	Compare with IATF 16949:2016 section 8.5.4.1 "Preservation of product"; also packaging instructions and specifications for technical cleanliness; product-specific cleanliness of the containers		☐	☐	☐	☒
7.14 P	Special characteristics: Are the special characteristics marked in the production control plan and are they systematically monitored in the production process.	Quality records for special characteristics have been defined. Marking of parts according to GS 91001. Inspection and control mechanisms according to the control plan for special characteristics may only be changed by trained personnel and shall be protected by a release hierarchy (incl. 4-eye principle) against unauthorized changes in the respective workstation.		☐	☐	☐	☒

No.	Question	Explanations	Comment	Evaluation			Not relevant
				RED	YELLOW	GREEN	
7.15 P T	Handling of faulty products: Has the procedure for handling faulty products been implemented in the process?	Marking of parts and areas, quarantine storage / containers, rework areas, retest, rework and marking of reworked parts; if required unsolicited (fault) notification, deviation permits, Further use of defective products IATF 16949:2016 section 8.7.1.7		☐	☐	☐	☒
7.16 P	Treatment of faulty products: Are reworks and re-checks conducted in accordance with work instructions and test instructions?	Lab Technology: Are the tests conducted according to the valid specifications?		☐	☐	☐	☒
7.17 P	Capacities: Is the OEE regularly reviewed and are measures for target attainment defined and followed up?			☐	☐	☐	☒
7.18 P	Logistics: Are the customer specifications for the logistical requirements known, evaluated and implemented?	Customer-specific requirements of the BMW Group (e.g. packaging manual)		☐	☐	☐	☒
7.19 P	Verification of system functionality: Is it ensured that after a planned or unplanned interruption of the manufacturing process (system) a test run with „red rabbits“ is performed in order to detect faulty samples and reject these?			☐	☐	☐	☒
7.20 P T	Product certification and marking of part: Are all product certificates required for the part still valid? Does the part have the required markings and do these correspond to the specifications on the product certificates and the drawing?	The validity of product certificates e.g. in China is tied to a successful annual plant audit of the Chinese authorities. The successfully performed audit shall be verified by means of a protocol/confirmation. Mandatory part markings (e.g. CCC label, model name, ECE approval mark) shall exist on the part and correspond to the specifications on the product certificates and the drawing.		☐	☐	☐	☒

No. P T	Question	Explanations	Comment	Evaluation			Not relevant
				RED	YELLOW	GREEN	
8	Maintenance						
8.1 P T	Maintenance: Is a preventive maintenance plan available and is the maintenance plan complied with (incl. documentation in the machine/tool logbook)?	e.g. Computer-aided system for planned maintenance		☐	☐	☐	☒
8.2 T	Maintenance: Are plans available for the spare part supply for production facilities and resources?	List of spare parts, spare parts warehouse, lead times, maintenance contracts, risk assessment, contingency plan		☐	☐	☐	☒
8.3 T	Maintenance: Is maintenance personnel available in case of system faults? How is the shift operation secured?	e.g. Response times, availability		☐	☐	☐	☒

No.	Question	Explanations	Comment	Evaluation			Not relevant
				RED	YELLOW	GREEN	
P T							
9	Test equipment						
9.1 P	Test equipment: Is all the required test equipment installed in the production and checked according to GS 98000?	Lab Technology: Are measurement system analysis tests conducted and documented? Is there an established procedure for measurement system analysis? Certification of test equipment in accordance with IATF 16949:2016 section 7.1.5.3.2 "External laboratories"		☐	☐	☐	☒
9.2 P T	Test equipment monitoring: Has test equipment monitoring been defined?	Planning, monitoring, identification, maintenance		☐	☐	☐	☒
9.3 P	E/E further development: Is the component and/or subsystem test station up to date?			☐	☐	☐	☒

P = practice
T = theory

**BMW
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Measures sheet	
Requalification product and process in accordance with GS 90018-2	Date:

Project:	Supplier:
Description:	Production interval:
Part number:	No. of items:

Participant BMW	QMT:	Participant supplier:	
	Logistics:		
	Purchasing:		
	Development:		
	Testing:		
	Other:		

[illegible]